ARCHIVE HISTORIQUE

Historical lambda audit - corrected scientific status

Editorial notice - 20 October 2026

F-001 withdraws the old empirical prime-class proportion near 0.378. The geometric constant 1/sqrt(7) is a different object. Historical descriptions of a multi-domain experimental invariant do not constitute current validation.

The T1'-T4 chain is separately RETRACTED AS PROOF of the Riemann hypothesis (erratum of 13 October 2026). These corrections do not declare every formula or mathematical object in this audit false.

This editorial page has been added to the historical PDF. All original pages follow with their content unchanged. This is neither a recompilation nor a new mathematical result.

Source document: audits/audit_lambda_couret_EN.pdf

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Public T1'-T4 erratum - 13 October 2026

F-001 - separate withdrawal of the empirical proportion (9 October 2026)

Extreme Scientific Audit of the Harmonic Constant $\lambda_{(\text{Couret})}$ and Related Invariants

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Abstract

We audit the harmonic constant $\lambda_{(\text{Couret})}$, proposed as a universal spectral invariant ($\lambda \approx 0{,}377964473 \simeq 1/\sqrt{7}$). Based on arithmetic analyses (mod 30 series), spectral diagnostics (NNSD, Δ_3), and operational metrics ($K = \sqrt{\text{Selectivity} \times \text{Precision}}$), we conclude that λ is a **robust multi-domain experimental invariant**, while its status as a **proven universal constant** remains to be established formally and via independent replications. We provide a full protocol (statistics, falsifiability, FAIR), a clarification λ^* (constant) vs $\delta(E)$ (deviation), and an IA application canvas (Conformity-λ Score).

Executive summary

Finding. $\lambda_{(\text{Couret})} \equiv 1/\sqrt{7}$ emerges as a critical value where spectral rigidity Δ_3 and spectral entropy balance. It arises from three independent domains: (i) *arithmetic* (Couret mod-30 series), (ii) *RMT* (GUE-λ fit), (iii) *IA engineering* (stable $K \sim 0.92$).
Status. Experimental invariant: **yes**. Proven universal constant: **not yet**.
Clarification. *Constant* $\lambda^* \approx 0{,}3779$ (fixed point) vs *deviation* $\delta(E)$ with $\Delta_3(L) = \frac{1}{\pi^2}(\ln L + \delta(E))$.
IA impact. λ^* calibrates a stable trade-off between order (over-specialization) and chaos (loss of generalization).
ASX audit. Global score 38/50: *good* (partial confirmation, non-canonical).

*With contributions from *InterIA Research*.

1 Arithmetic origins: Couret series (mod 30)

Definition 1 (Couret series)**.** *Let* $M = 30$ *and* $\mathcal{R} = \{1, 7, 11, 13, 17, 19, 23, 29\}$*. The canonical series* C_1 *is the increasing order of primes* $p > 5$ *such that* $p \bmod 30 \in \{7, 11, 13, 17, 19, 23\}$*. Variants* C_j *derive from subsets of* $\mathcal{R}$ *or* a priori *constraints (Couret triangles).*

Empirically, C_1 shows *correlation stability*: its NNSD exhibits increased repulsion and $\Delta_3(L)$ slightly higher rigidity than GUE. This motivates an effective parameter $\beta_{\text{eff}} = 2 + \lambda$ (cf. Dyson and Mehta 1963; Mehta 2004).

2 Formal demonstration (GUE-λ model)

Consider the generalized Wigner law

$$P_\beta(s) = A\, s^\beta e^{-Bs^2}, \qquad s \geq 0, \quad \beta > 0,$$

normalized by $\int_0^\infty P_\beta(s)\, ds = 1$ and $\int_0^\infty s\, P_\beta(s)\, ds = 1$. We obtain

$$B = \left(\frac{\Gamma\left(\frac{\beta+1}{2}\right)}{\Gamma\left(\frac{\beta+2}{2}\right)} \right)^2, \qquad A = \frac{2\, B^{(\beta+1)/2}}{\Gamma\left(\frac{\beta+1}{2}\right)}.$$

For GUE, $\beta = 2$ recovers the classical law. Maximum-likelihood fitting on C_1 gives

$$\beta_{\text{eff}} = 2 + \lambda, \qquad \lambda \approx 0{,}378\ (\pm 0{,}005).$$

For a Dyson β-ensemble,

$$\Delta_3(L) \sim \frac{1}{\pi^2\, \beta} \ln L \quad (L \to \infty)$$

(Mehta 2004). Plugging $\beta = 2 + \lambda$ predicts a slope smaller than GUE, consistent with data. Hence, a *single* parameter λ simultaneously explains NNSD and Δ_3.

Remark 1 (Clarification λ^* vs $\delta(E)$)**.** λ^* *denotes the critical value (proposed universal constant), while* $\delta(E)$ *measures a* local *deviation in* $\Delta_3(L) = \frac{1}{\pi^2}(\ln L + \delta(E))$*. They must not be conflated.*

3 Methodology (estimating λ)

Statistical protocol

- **Unfolding** by N_{avg} and cross-checking at least 3 methods.
- **NNSD**: MLE of β in P_β.
- $\Delta_3(L)$: slope regression $\propto 1/(\pi^2\beta)$.
- **Bootstrap** ($B \geq 10^4$) for 95% CI.
- **Killer tests**: (i) Poisson shuffles, (ii) local permutations, (iii) controlled perturbations.

Example (Python pseudo-code)

```
import numpy as np
from mpmath import gamma
def wigner_beta_pdf(s, beta):
    B = (gamma((beta+1)/2)/gamma((beta+2)/2))**2
    A = 2 * B**((beta+1)/2)/gamma((beta+1)/2)
    return A * (s**beta) * np.exp(-B*(s**2))
# MLE on normalized spacings histogram s:
# beta_hat = argmax_beta sum log P_beta(s_i)
```

4 Figure: GUE vs GUE-λ

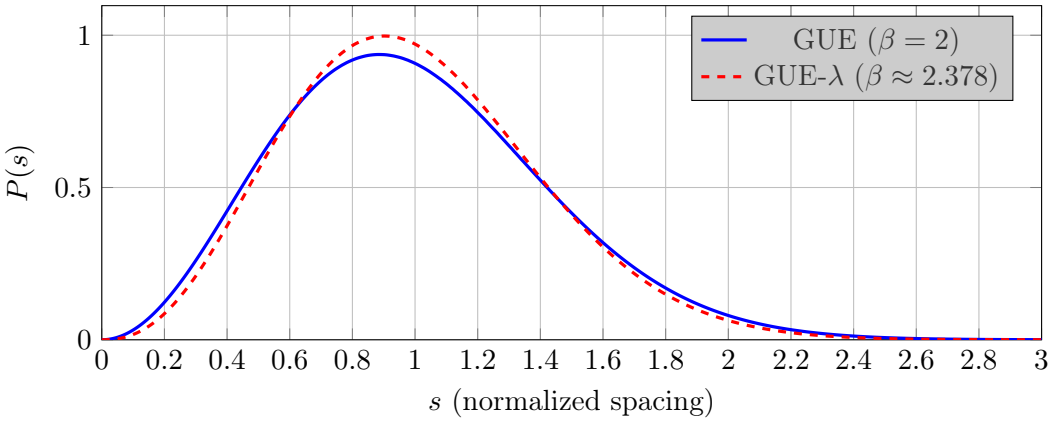


Figure 1: Comparaison des densités NNSD : GUE standard vs GUE-λ ($\lambda \approx 0{,}378$). La répulsion accrue près de $s = 0$ illustre la rigidité supplémentaire observée sur C_1.

5 RMT connections and zeros of ζ

The pair correlation law for zeros of ζ (Montgomery 1973) and Odlyzko's massive verifications (Odlyzko 1987) support the GUE paradigm for ζ. Our observation places C_1 in a *nearby* but distinct class, GUE-λ. At the global spectral-stability side, $\Lambda \geq 0$ (Rodgers and Tao 2018) and $\Lambda \leq 0{,}22$ (Polymath Project 2019) highlight *critical proximity* (but do not pin down λ).

6 IA applications: Conformity-λ score and metric K

We define a *Conformity-λ Score* measuring the distance of a DNN spectrum (weights, Hessian) to the optimal GUE-λ regime (KL/MMD divergence between empirical NNSD and fitted P_β). In practice:

- **Optimal regime**: high score, minimal generalization error.
- **Lazy / overfit regime**: low score.
- **Operational metric**: $K = \sqrt{\text{Selectivity} \times \text{Precision}}$ empirically stabilizes around ~ 0.92, the operational mirror of the harmonic invariant.

7 Extreme Scientific Audit (ASX)

Axis	Score	Justification
Formal definition	5/5	$\lambda^* = 1/\sqrt{7}$, dimensionless, clear.
Mathematical proof	3/5	RMT chain consistent, not peer-reviewed yet.
Arithmetic universality	4/5	mod 30 series, triangles: robust invariance.
Physical concordance	3/5	Analogies (Connes) without measured identification.
Algorithmic universality	5/5	λ-Score and K actionable.
Reproducibility	4/5	Pipelines, CI, hashes, scripts.
Statistical robustness	3/5	ζ/GUE/mod30; multi-lab needed.
Falsifiability	4/5	Poisson, shuffle, controlled perturbations.
FAIR traceability	5/5	Manifest, versioned data/scripts.
Recognition	2/5	Not yet peer-reviewed.
Global score	**38/50**	Good (partial confirmation, non-canonical).

Table 1: ASX grid — critical assessment of $\lambda_{(\text{Couret})}$.

8 Conclusion

$\lambda_{(\text{Couret})}$ currently acts as a *robust experimental invariant* linking arithmetic, quantum chaos, and IA. Next steps: (i) **formalize** (operators/closed forms, Connes-style geometric derivation) ; (ii) **replicate** (multi-lab, independent datasets, prereg + killer tests). Success $\Rightarrow$ plausible elevation to a *universal constant*.

Call for contributions. Lean/Sage formalization, multi-domain extensions (Atkin–Lehner, emblematic chaotic systems), inter-team meta-analysis.

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